

JACKSON MORRIS

CURRICULUM VITAE

Simon Laufer Mathematical Science Institute

Berkeley, CA 94720 | Office 303

jacksonmorris1999@gmail.com | [jackson-morris.github.io](https://github.com/jackson-morris)

EMPLOYMENT

Postdoctoral Researcher

Vanderbilt University

2027-2029

Postdoc SLMath

SLMath Semester Program

Motivic Homotopy Theory: Connections and Applications

Fall 2026

EDUCATION

Ph.D. in Mathematics

University of Washington

Advisors: Kyle Ormsby and John Palmieri

Thesis title: Cooperations in motivic homotopy theory

2021-2026

M.S. in Mathematics

University of Washington

2021-2023

B.S. in Mathematics, *magna cum laude*

University of Kentucky

2017-2020

RESEARCH INTERESTS

Motivic homotopy theory, chromatic homotopy theory, hermitian K-theory, algebraic K-theory, splittings, synthetic spectra, nilpotence and periodicity, Adams spectral sequence, periodic self-maps, redshift and blueshift

PUBLICATIONS AND PREPRINTS

10. *C₂-equivariant C-motivic stable stems*, joint with J.D. Quigley. In preparation.
9. *Diachromatic blueshift*, joint with Kyle Ormsby. In preparation.
8. *Motivic K-theory cooperations over the rationals*, joint with Sarah Petersen and Liz Tatum. In preparation.
7. *An exotic self-map of periodicity 1*. In preparation.
6. *Periodic phenomena in stable motivic homotopy theory*. Submitted. Available at <https://arxiv.org/abs/2607.18165>
5. *Immersion of C₂-projective spaces via Kℝ-theory*, joint with Manyi Guo, Alex Waugh, and Albert Jinghui Yang. Submitted. Available at <https://arxiv.org/abs/2604.25260>.
4. *Spectrum level splittings of truncated motivic Brown-Peterson cooperations algebras*, joint with Sarah Petersen and Elizabeth Tatum. Submitted. Available at <https://arxiv.org/abs/2509.19542>.
3. *Rings of cooperations for hermitian K-theory over finite fields*. To appear in **Algebraic & Geometric Topology**. Available at <https://arxiv.org/abs/2509.02786>.
2. *On the ring of cooperations for real hermitian K-theory*. To appear in **Annals of K-theory**. Available at <https://arxiv.org/abs/2506.16672>.
1. *Toric Double Determinantal Varieties*, joint with Alexander Blose, Patricia Klein, and Owen McGrath. **Communications in Algebra** 49 (2020), 7, 3085-3093.

TEACHING EXPERIENCE

Instructor of Record

University of Washington

Math 480 – Homotopy theory

Spring 2026

Math 208 – Linear algebra

Summer 2023

Math 208 – Linear algebra

Winter 2023

Graduate Assistant

Electronic Computational Homotopy Theory (eCHT)

Research Seminar

2025–2026

Quadratic Curve Counting

Fall 2024

Park City Math Institute (PCMI)

Summer 2024

Experimental Math Lab

University of Washington

2021–2026

Courses included: Precalculus, Differential calculus, Multivariable calculus, Linear algebra, Introduction to proofs, Topology

Advising/Mentoring

Directed Reading Mentor

UW

Topics included: Homotopy theory, Representation theory, Geometric group theory, Homological algebra

2022–2025

Advising/Mentorship

Jason Yeung (University of Toronto)

SLMath

Program associate mentor during thematic program

Fall 2026

Spencer Catron (University of Washington)

UW

Graduate student mentor during first year

2023–2024

Math Olympiad Judge

UW Math Hour

2023–2024

Judge for elementary school math competition

HONORS AND AWARDS

Lee and Cozette McFarlan Travel Fellowship

Summer 2025

University of Washington

\$750

For travel to European Talbot in Kolding, DK.

Research Assistantship

Summer 2025

University of Washington

For good research progress.

JC Eaves Scholarship

2020

University of Kentucky

\$2000

For continued excellence in mathematics as a Kentuckian.

Carolyn Bunyan Award

2020

University of Kentucky

\$2000

For mathematics majors interested in continuing education in mathematics.

SERVICE AND PROFESSIONAL ACTIVITIES

Refereeing and Reviewing

Algebraic and Geometric Topology

2025–Present

ZBMath

2026–Present

Seminar Organization

DUBTOP Seminar

UW

Topics included: Chromatic homotopy theory, stacks, Goodwillie calculus

2024–2025

Back-To-School seminar

Student lectures on fun math learned over the summer

UW
Fall 2024**Reading seminars***Topics included:* Algebraic K-theory, cohomology operations, motivic homotopy theory, vector bundles and topological K-theory, categorical homotopy theory, model categoriesUW
2023-2024**SELECTED PRESENTATIONS & SEMINAR TALKS**

Conference and Colloquium Talks**AMS Sectional: Computational Homotopy Theory***An exotic self-map of periodicity 1*Tempe, AZ
November 2026**Motivic Homotopy Theory: Connections and Applications***Periodicity in motivic homotopy theory*SLMath, Berkeley, CA
October 2026**Young Topologists Meeting***Higher Witt K-theories*Copenhagen, DK
July 2026**Cascade Topology Seminar***Higher Witt K-theories*Portland, OR
May 2026**Joint Mathematics Meetings 2026***Splittings and cooperations in motivic homotopy theory*Washington, D.C.
January 2026**Graduate Student Geometry & Topology Seminar***On the ring of cooperations for real hermitian K-theory*Bloomington, IN
April 2025**Reed Mathematics Colloquium***Finding Fixed Points with Invariants*Portland, OR
February 2025**Joint Mathematics Meetings 2025***On the ring of cooperations for real hermitian K-theory*Seattle, WA
January 2025Seminar Talks**New Mexico State University Topology Seminar***Higher Witt K-theories*Las Cruces, NM
September 2026**eCHT Research Seminar***Higher Witt K-theories*Online
September 2026**University of Washington Topology Seminar***Spectrum level splittings of truncated motivic Brown-Peterson cooperations algebras*Seattle, WA
February 2026**University of Illinois Urbana-Champaign Topology Seminar***Splittings and cooperations in motivic homotopy theory*Champaign, IL
January 2026**University of Virginia Topology Seminar***Splittings and cooperations in motivic homotopy theory*Charlottesville, VA
November 2025**University of Kentucky Topology Seminar***Splittings and cooperations in motivic homotopy theory*Lexington, KY
November 2025**Duke University Geometry and Topology Seminar***Splittings and cooperations in motivic homotopy theory*Durham, NC
October 2025**University of Colorado Boulder Homotopy Theory Seminar***Spectrum level splittings of truncated motivic Brown-Peterson cooperations algebras*Boulder, CO
September 2025**University of Michigan Geometry Seminar***On the ring of cooperations for real hermitian K-theory*Ann Arbor, MI
September 2025**University of Notre Dame Topology Seminar***Splittings and cooperations in motivic homotopy theory*South Bend, IN
September 2025**University of Washington Topology Seminar***On the ring of cooperations for real hermitian K-theory*Seattle, WA
June 2025Contributed Talks

Reading Seminars

An upper bound on the degree of symmetry of an exotic sphere
Higher Adams differentials and hidden extensions
Motivic twisted K-theory
Higher real K-theories, redshift, and blueshift
Filtered objects, spectral sequences, and t-structures
(A synthetic approach to detecting) v_1 -periodic families
Algebraic and Milnor K-theory
Galois descent and the Picard group of K-theory
The Adams spectral sequence and Hopf algebroids
Symplectic orientations

March 2026
November 2025
April 2024
May 2026
January 2026
November 2025
October 2025
May 2025
February 2025
August 2024

Workshops

$\pi_* THH(\mathbb{Z}_p^{fil})/p$ and $\pi_* THH(\ell_p^{fil})/(p, v_1)$ (Talbot, Nacogdoches TX)
Hochschild Homology (European Autumn School in Topology, Utrecht, NL)

August 2024
September 2024